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Editorial Office
Physica A: Statistical Mechanics and its Applications

Dear Editor,

I am submitting the manuscript "A Geometric Observatory for Financial Regime Detection" for consideration in Physica A.

The paper applies differential geometry—Fubini–Study metrics, Berry curvature, Chern numbers, spectral gaps—to financial time series by embedding market data into a projective Hilbert space via spectral metric learning (QCML). We extract nineteen geometric detection channels and test them on 17 historical crises spanning 2000–2024, comparing against 46 methods including Random Forest, GARCH, Hamilton Markov-switching, and Absorption Ratio.

The central result: walk-forward nested HPO gives Berry Phase Rate an unbiased out-of-sample Cohen's $d = 0.72$ with 30% fewer false alarms than Random Forest (2.5 vs. 3.6 per year), without requiring crisis labels. Regime-Adaptive fusion generalizes on holdout crises ($d = 0.78$) while top individual methods collapse.

I believe this manuscript is a good fit for Physica A because it brings geometric and spectral methods from mathematical physics to an econophysics application, with rigorous statistical evaluation. The work includes formal theorems validated on market data, a 46-method comparison with proper multiple-testing corrections, and walk-forward validation with nested hyperparameter optimization.

All code and data are freely available for reproducibility at `https://github.com/willhammondhimself/qcml-geometric-sde`.

Sincerely,

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